Final Assignment – FinTech Course

The final assignment requires students to address a given problem starting from the assigned dataset.

The delivery must include a well-structured code implementation (in a Jupyter Notebook or Google Colab Notebook). The code should be clearly commented, or alternatively accompanied by a written report describing the methodology, implementation, and results.

In addition, students are required to prepare a PowerPoint presentation that frames the problem, describe the dataset, explains the approach taken, highlights the main results, and concludes with key insights and considerations.

Your assignment:

Default of payments (Dataset 3)

This dataset contains information on default payments, demographic factors, credit data, history of payment, and bill statements of credit card clients in Taiwan from April 2005 to September 2005.

Content

There are 25 variables:

1. ID: ID of each client
2. LIMIT_BAL: Amount of given credit in NT dollars (includes individual and family/supplementary credit
3. SEX: Gender (1=male, 2=female)
4. EDUCATION: (1=graduate school, 2=university, 3=high school, 4=others, 5=unknown, 6=unknown)
5. MARRIAGE: Marital status (1=married, 2=single, 3=others)
6. AGE: Age in years
7. PAY_0: Repayment status in September, 2005 (-1=pay duly, 1=payment delay for one month, 2=payment delay for two months, … 8=payment delay for eight months, 9=payment delay for nine months and above)
8. PAY_2: Repayment status in August, 2005 (scale same as above)
9. PAY_3: Repayment status in July, 2005 (scale same as above)
10. PAY_4: Repayment status in June, 2005 (scale same as above)
11. PAY_5: Repayment status in May, 2005 (scale same as above)
12. PAY_6: Repayment status in April, 2005 (scale same as above)
13. BILL_AMT1: Amount of bill statement in September, 2005 (NT dollar)
14. BILL_AMT2: Amount of bill statement in August, 2005 (NT dollar)
15. BILL_AMT3: Amount of bill statement in July, 2005 (NT dollar)
16. BILL_AMT4: Amount of bill statement in June, 2005 (NT dollar)
17. BILL_AMT5: Amount of bill statement in May, 2005 (NT dollar)
18. BILL_AMT6: Amount of bill statement in April, 2005 (NT dollar)
19. PAY_AMT1: Amount of previous payment in September, 2005 (NT dollar)
20. PAY_AMT2: Amount of previous payment in August, 2005 (NT dollar)
21. PAY_AMT3: Amount of previous payment in July, 2005 (NT dollar)
22. PAY_AMT4: Amount of previous payment in June, 2005 (NT dollar)
23. PAY_AMT5: Amount of previous payment in May, 2005 (NT dollar)
24. PAY_AMT6: Amount of previous payment in April, 2005 (NT dollar)
25. default.payment.next.month: Default payment (1=yes, 0=no)

What to do:

- *propose a ML classifier exploiting a random forest and another with XGBoost;*
- *now repeat the above exercise*

- *now repaat the previous exercise AFTER exploiting a Lasso regression to reduce the number of features*

and compare the two classifiers. Always choose the optimal hyperparameters. What are the most important drivers of the default?

Moreover, is the dataset unbalanced? If yes, investigate the issue and propose a solution.